R Bootcamp - Part II

An Introduction to R Programming for Public Health Students

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Contents

1 Loading Data

We will use the *Number of Reported Malaria Cases by County— United States, 2017* dataset available at data.cdc.gov/Global-Health/Number-of-Reported-Malaria-Cases-by-County-United-/puzh-5wax/about_data. Download the CSV file onto your computer to be able to easily work with it.

```
# Load a CSV file
data_malaria = read.csv(
  "Data/Number_of_Reported_Malaria_Cases_by_County__United_States__2017_20250120.csv"
  )
# Notice the changes in the Environmental pane

# Preview the data
head(data_malaria)
```

```
##         COUNTRY      STATE   COUNTY MAL_FREQ_2017 STATE_CODE_FIPS
## 1 UNITED STATES    ARIZONA MARICOPA            20               4
## 2 UNITED STATES   COLORADO ARAPAHOE             8               8
## 3 UNITED STATES   COLORADO   DENVER             6               8
```

```
## 4 UNITED STATES CONNECTICUT FAIRFIELD             5                9
## 5 UNITED STATES CONNECTICUT  HARTFORD            10                9
## 6 UNITED STATES CONNECTICUT NEW HAVEN             7                9
##   COUNTY_CODE_FIPS FIPS
## 1               13 4013
## 2                5 8005
## 3               31 8031
## 4                1 9001
## 5                3 9003
## 6                9 9009
```

2 Exploring Data (continued)

We can explore and summarize data with functions like `summary()`, `unique()`, `table()`, `aggregate()`...

Data exploration helps understand the dataset structure and key statistics:

```r
# Example: Summary statistics
summary(data_malaria)
```

```
##    COUNTRY              STATE              COUNTY          MAL_FREQ_2017
##  Length:97          Length:97          Length:97          Min.   :  5.0
##  Class :character   Class :character   Class :character   1st Qu.:  6.0
##  Mode  :character   Mode  :character   Mode  :character   Median :  8.0
##                                                           Mean   : 14.6
##                                                           3rd Qu.: 19.0
##                                                           Max.   :101.0
##  STATE_CODE_FIPS COUNTY_CODE_FIPS      FIPS
##  Min.   : 4.00   Min.   :  1.00   Min.   : 4013
##  1st Qu.:24.00   1st Qu.: 17.00   1st Qu.:24003
##  Median :34.00   Median : 53.00   Median :34013
##  Mean   :30.68   Mean   : 74.98   Mean   :30755
##  3rd Qu.:39.00   3rd Qu.: 99.00   3rd Qu.:39061
##  Max.   :53.00   Max.   :510.00   Max.   :53061
```

```r
# Find unique values
unique(data_malaria$STATE)
```

```
##  [1] "ARIZONA"              "COLORADO"             "CONNECTICUT"
##  [4] "DISTRICT OF COLUMBIA" "FLORIDA"              "GEORGIA"
##  [7] "ILLINOIS"             "INDIANA"              "IOWA"
## [10] "KANSAS"               "MAINE"                "MARYLAND"
## [13] "MASSACHUSETTS"        "MICHIGAN"             "MINNESOTA"
## [16] "MISSOURI"             "NEW HAMPSHIRE"        "NEW JERSEY"
## [19] "NEW YORK"             "NORTH CAROLINA"       "NORTH DAKOTA"
## [22] "OHIO"                 "OREGON"               "PENNSYLVANIA"
## [25] "RHODE ISLAND"         "SOUTH DAKOTA"         "TEXAS"
## [28] "VIRGINIA"             "WASHINGTON"
```

```r
# Create a contingency table
table(data_malaria$STATE) # Number of observations by state
```

```
##
##              ARIZONA             COLORADO          CONNECTICUT
##                    1                    2                    3
## DISTRICT OF COLUMBIA              FLORIDA              GEORGIA
##                    1                    6                    5
##             ILLINOIS              INDIANA                 IOWA
##                    1                    1                    1
##               KANSAS                MAINE             MARYLAND
##                    2                    1                    7
##        MASSACHUSETTS             MICHIGAN            MINNESOTA
##                    5                    5                    3
##             MISSOURI        NEW HAMPSHIRE           NEW JERSEY
##                    1                    1                    7
##             NEW YORK       NORTH CAROLINA         NORTH DAKOTA
##                   12                    4                    1
##                 OHIO               OREGON         PENNSYLVANIA
##                    4                    1                    5
##         RHODE ISLAND         SOUTH DAKOTA                TEXAS
##                    1                    1                    8
##             VIRGINIA           WASHINGTON
##                    5                    2
```

```
# Aggregate scores by passing status
aggregate(MAL_FREQ_2017 ~ STATE, data = data_malaria, FUN = sum)
```

```
##                   STATE MAL_FREQ_2017
## 1               ARIZONA            20
## 2              COLORADO            14
## 3           CONNECTICUT            22
## 4  DISTRICT OF COLUMBIA            19
## 5               FLORIDA            41
## 6               GEORGIA            59
## 7              ILLINOIS            29
## 8               INDIANA             9
## 9                  IOWA             7
## 10               KANSAS            11
## 11                MAINE            11
## 12             MARYLAND           237
## 13        MASSACHUSETTS            80
## 14             MICHIGAN            34
## 15            MINNESOTA            44
## 16             MISSOURI             5
## 17        NEW HAMPSHIRE             6
## 18           NEW JERSEY           103
## 19             NEW YORK           280
## 20       NORTH CAROLINA            33
## 21         NORTH DAKOTA             6
## 22                 OHIO            44
## 23               OREGON             7
## 24         PENNSYLVANIA            68
## 25         RHODE ISLAND            10
## 26         SOUTH DAKOTA             6
## 27                TEXAS           123
## 28             VIRGINIA            61
```

```
## 29           WASHINGTON            27
```

```r
aggregate(MAL_FREQ_2017 ~ STATE, data = data_malaria, FUN = sd)
```

```
##                   STATE MAL_FREQ_2017
## 1               ARIZONA            NA
## 2              COLORADO     1.4142136
## 3           CONNECTICUT     2.5166115
## 4  DISTRICT OF COLUMBIA            NA
## 5               FLORIDA     1.6020820
## 6               GEORGIA     7.5630682
## 7              ILLINOIS            NA
## 8               INDIANA            NA
## 9                  IOWA            NA
## 10               KANSAS     0.7071068
## 11                MAINE            NA
## 12             MARYLAND    36.7850901
## 13        MASSACHUSETTS    12.1655251
## 14             MICHIGAN     1.7888544
## 15            MINNESOTA    15.8850034
## 16             MISSOURI            NA
## 17        NEW HAMPSHIRE            NA
## 18           NEW JERSEY    10.5785047
## 19             NEW YORK    26.3760681
## 20       NORTH CAROLINA     1.7078251
## 21         NORTH DAKOTA            NA
## 22                 OHIO     5.9441848
## 23               OREGON            NA
## 24         PENNSYLVANIA    11.0589330
## 25         RHODE ISLAND            NA
## 26         SOUTH DAKOTA            NA
## 27                TEXAS     9.5907321
## 28             VIRGINIA     7.3620649
## 29           WASHINGTON    10.6066017
```

2.1 Activity

- What does the `aggregate()` function do?
- Calculate the mean and standard deviation of malaria cases for each state.
- Create a new data frame containing the information of the number of cases by state.

3 Subsetting Data

Subsetting helps in extracting specific rows, columns, or elements. It can be done with logical tests (`>`, `<`, `==`, `is.na()`) and operators (`!`, `&`, `|`).

```r
# Select one element
data_malaria[data_malaria$STATE == "COLORADO",  ]
```

```
##         COUNTRY    STATE   COUNTY MAL_FREQ_2017 STATE_CODE_FIPS
```

```
## 2 UNITED STATES COLORADO ARAPAHOE             8               8
## 3 UNITED STATES COLORADO   DENVER             6               8
##   COUNTY_CODE_FIPS FIPS
## 2                5 8005
## 3               31 8031
```

```
# Exclude one element
data_malaria[data_malaria$STATE != "NEW YORK",  ]
```

```
##          COUNTRY                STATE               COUNTY MAL_FREQ_2017
## 1  UNITED STATES              ARIZONA             MARICOPA            20
## 2  UNITED STATES             COLORADO             ARAPAHOE             8
## 3  UNITED STATES             COLORADO               DENVER             6
## 4  UNITED STATES          CONNECTICUT            FAIRFIELD             5
## 5  UNITED STATES          CONNECTICUT             HARTFORD            10
## 6  UNITED STATES          CONNECTICUT            NEW HAVEN             7
## 7  UNITED STATES DISTRICT OF COLUMBIA DISTRICT OF COLUMBIA            19
## 8  UNITED STATES              FLORIDA              BROWARD             8
## 9  UNITED STATES              FLORIDA                 DADE             9
## 10 UNITED STATES              FLORIDA         HILLSBOROUGH             7
## 11 UNITED STATES              FLORIDA                 LEON             7
## 12 UNITED STATES              FLORIDA              OSCEOLA             5
## 13 UNITED STATES              FLORIDA           PALM BEACH             5
## 14 UNITED STATES              GEORGIA              CLAYTON             5
## 15 UNITED STATES              GEORGIA                 COBB             8
## 16 UNITED STATES              GEORGIA               DEKALB            20
## 17 UNITED STATES              GEORGIA               FULTON             6
## 18 UNITED STATES              GEORGIA             GWINNETT            20
## 19 UNITED STATES             ILLINOIS                 COOK            29
## 20 UNITED STATES              INDIANA               MARION             9
## 21 UNITED STATES                 IOWA                 POLK             7
## 22 UNITED STATES               KANSAS              JOHNSON             5
## 23 UNITED STATES               KANSAS             SEDGWICK             6
## 24 UNITED STATES                MAINE           CUMBERLAND            11
## 25 UNITED STATES             MARYLAND         ANNE ARUNDEL            10
## 26 UNITED STATES             MARYLAND            BALTIMORE            23
## 27 UNITED STATES             MARYLAND            FREDERICK             6
## 28 UNITED STATES             MARYLAND               HOWARD            11
## 29 UNITED STATES             MARYLAND           MONTGOMERY            70
## 30 UNITED STATES             MARYLAND        PRINCE GEORGE           101
## 31 UNITED STATES             MARYLAND       BALTIMORE CITY            16
## 32 UNITED STATES        MASSACHUSETTS              HAMPDEN             5
## 33 UNITED STATES        MASSACHUSETTS            MIDDLESEX            34
## 34 UNITED STATES        MASSACHUSETTS             PLYMOUTH             5
## 35 UNITED STATES        MASSACHUSETTS              SUFFOLK            15
## 36 UNITED STATES        MASSACHUSETTS            WORCESTER            21
## 37 UNITED STATES             MICHIGAN               INGHAM             5
## 38 UNITED STATES             MICHIGAN                 KENT             9
## 39 UNITED STATES             MICHIGAN              OAKLAND             7
## 40 UNITED STATES             MICHIGAN            WASHTENAW             5
## 41 UNITED STATES             MICHIGAN                WAYNE             8
## 42 UNITED STATES            MINNESOTA                ANOKA             5
## 43 UNITED STATES            MINNESOTA             HENNEPIN            33
## 44 UNITED STATES            MINNESOTA               RAMSEY             6
```

```
## 45 UNITED STATES           MISSOURI           ST. LOUIS             5
## 46 UNITED STATES      NEW HAMPSHIRE         HILLSBOROUGH             6
## 47 UNITED STATES         NEW JERSEY              BERGEN             8
## 48 UNITED STATES         NEW JERSEY          BURLINGTON             8
## 49 UNITED STATES         NEW JERSEY               ESSEX            37
## 50 UNITED STATES         NEW JERSEY              HUDSON            12
## 51 UNITED STATES         NEW JERSEY              MERCER            11
## 52 UNITED STATES         NEW JERSEY           MIDDLESEX            19
## 53 UNITED STATES         NEW JERSEY               UNION             8
## 66 UNITED STATES     NORTH CAROLINA              DURHAM             8
## 67 UNITED STATES     NORTH CAROLINA            GUILFORD             9
## 68 UNITED STATES     NORTH CAROLINA         MECKLENBURG             6
## 69 UNITED STATES     NORTH CAROLINA                WAKE            10
## 70 UNITED STATES       NORTH DAKOTA                CASS             6
## 71 UNITED STATES               OHIO            CUYAHOGA            12
## 72 UNITED STATES               OHIO            FRANKLIN            19
## 73 UNITED STATES               OHIO            HAMILTON             6
## 74 UNITED STATES               OHIO          MONTGOMERY             7
## 75 UNITED STATES             OREGON            MULTNOMAH             7
## 76 UNITED STATES       PENNSYLVANIA           ALLEGHENY             8
## 77 UNITED STATES       PENNSYLVANIA               BERKS             5
## 78 UNITED STATES       PENNSYLVANIA             DAUPHIN             5
## 79 UNITED STATES       PENNSYLVANIA            DELAWARE            20
## 80 UNITED STATES       PENNSYLVANIA        PHILADELPHIA            30
## 81 UNITED STATES       RHODE ISLAND          PROVIDENCE            10
## 82 UNITED STATES       SOUTH DAKOTA           MINNEHAHA             6
## 83 UNITED STATES              TEXAS              AUSTIN            12
## 84 UNITED STATES              TEXAS               BEXAR             5
## 85 UNITED STATES              TEXAS              COLLIN             7
## 86 UNITED STATES              TEXAS              DALLAS            19
## 87 UNITED STATES              TEXAS           FORT BEND            17
## 88 UNITED STATES              TEXAS              HARRIS             9
## 89 UNITED STATES              TEXAS             HOUSTON            35
## 90 UNITED STATES              TEXAS             TARRANT            19
## 91 UNITED STATES           VIRGINIA           ARLINGTON             5
## 92 UNITED STATES           VIRGINIA             FAIRFAX            24
## 93 UNITED STATES           VIRGINIA             LOUDOUN             8
## 94 UNITED STATES           VIRGINIA      PRINCE WILLIAM            14
## 95 UNITED STATES           VIRGINIA     ALEXANDRIA CITY            10
## 96 UNITED STATES         WASHINGTON                KING            21
## 97 UNITED STATES         WASHINGTON           SNOHOMISH             6
##    STATE_CODE_FIPS COUNTY_CODE_FIPS  FIPS
## 1                4               13  4013
## 2                8                5  8005
## 3                8               31  8031
## 4                9                1  9001
## 5                9                3  9003
## 6                9                9  9009
## 7               11                1 11001
## 8               12               11 12011
## 9               12               25 12025
## 10              12               57 12057
## 11              12               73 12073
## 12              12               97 12097
```

```
## 13                12              99  12099
## 14                13              63  13063
## 15                13              67  13067
## 16                13              89  13089
## 17                13             121  13121
## 18                13             135  13135
## 19                17              31  17031
## 20                18              97  18097
## 21                19             153  19153
## 22                20              91  20091
## 23                20             173  20173
## 24                23               5  23005
## 25                24               3  24003
## 26                24               5  24005
## 27                24              21  24021
## 28                24              27  24027
## 29                24              31  24031
## 30                24              33  24033
## 31                24             510  24510
## 32                25              13  25013
## 33                25              17  25017
## 34                25              23  25023
## 35                25              25  25025
## 36                25              27  25027
## 37                26              65  26065
## 38                26              81  26081
## 39                26             125  26125
## 40                26             161  26161
## 41                26             163  26163
## 42                27               3  27003
## 43                27              53  27053
## 44                27             123  27123
## 45                29             189  29189
## 46                33              11  33011
## 47                34               3  34003
## 48                34               5  34005
## 49                34              13  34013
## 50                34              17  34017
## 51                34              21  34021
## 52                34              23  34023
## 53                34              39  34039
## 66                37              63  37063
## 67                37              81  37081
## 68                37             119  37119
## 69                37             183  37183
## 70                38              17  38017
## 71                39              35  39035
## 72                39              49  39049
## 73                39              61  39061
## 74                39             113  39113
## 75                41              51  41051
## 76                42               3  42003
## 77                42              11  42011
## 78                42              43  42043
```

```
## 79             42             45 42045
## 80             42            101 42101
## 81             44              7 44007
## 82             46             99 46099
## 83             48             15 48015
## 84             48             29 48029
## 85             48             85 48085
## 86             48            113 48113
## 87             48            157 48157
## 88             48            201 48201
## 89             48            225 48225
## 90             48            439 48439
## 91             51             13 51013
## 92             51             59 51059
## 93             51            107 51107
## 94             51            153 51153
## 95             51            510 51510
## 96             53             33 53033
## 97             53             61 53061
```

```r
# Select values above a threshold
data_malaria[data_malaria$MAL_FREQ_2017 > 10,  ]
```

```
##          COUNTRY                STATE               COUNTY MAL_FREQ_2017
## 1  UNITED STATES              ARIZONA             MARICOPA            20
## 7  UNITED STATES DISTRICT OF COLUMBIA DISTRICT OF COLUMBIA            19
## 16 UNITED STATES              GEORGIA               DEKALB            20
## 18 UNITED STATES              GEORGIA             GWINNETT            20
## 19 UNITED STATES             ILLINOIS                 COOK            29
## 24 UNITED STATES                MAINE           CUMBERLAND            11
## 26 UNITED STATES             MARYLAND            BALTIMORE            23
## 28 UNITED STATES             MARYLAND               HOWARD            11
## 29 UNITED STATES             MARYLAND           MONTGOMERY            70
## 30 UNITED STATES             MARYLAND       PRINCE GEORGE           101
## 31 UNITED STATES             MARYLAND       BALTIMORE CITY            16
## 33 UNITED STATES        MASSACHUSETTS            MIDDLESEX            34
## 35 UNITED STATES        MASSACHUSETTS              SUFFOLK            15
## 36 UNITED STATES        MASSACHUSETTS            WORCESTER            21
## 43 UNITED STATES            MINNESOTA             HENNEPIN            33
## 49 UNITED STATES           NEW JERSEY                ESSEX            37
## 50 UNITED STATES           NEW JERSEY               HUDSON            12
## 51 UNITED STATES           NEW JERSEY               MERCER            11
## 52 UNITED STATES           NEW JERSEY            MIDDLESEX            19
## 54 UNITED STATES             NEW YORK               ALBANY            11
## 55 UNITED STATES             NEW YORK                BRONX            91
## 57 UNITED STATES             NEW YORK                KINGS            50
## 60 UNITED STATES             NEW YORK             NEW YORK            42
## 62 UNITED STATES             NEW YORK               QUEENS            33
## 63 UNITED STATES             NEW YORK             RICHMOND            11
## 71 UNITED STATES                 OHIO             CUYAHOGA            12
## 72 UNITED STATES                 OHIO             FRANKLIN            19
## 79 UNITED STATES         PENNSYLVANIA             DELAWARE            20
## 80 UNITED STATES         PENNSYLVANIA         PHILADELPHIA            30
## 83 UNITED STATES                TEXAS               AUSTIN            12
```

```
## 86 UNITED STATES                TEXAS            DALLAS            19
## 87 UNITED STATES                TEXAS         FORT BEND            17
## 89 UNITED STATES                TEXAS           HOUSTON            35
## 90 UNITED STATES                TEXAS           TARRANT            19
## 92 UNITED STATES             VIRGINIA           FAIRFAX            24
## 94 UNITED STATES             VIRGINIA    PRINCE WILLIAM            14
## 96 UNITED STATES           WASHINGTON              KING            21
##    STATE_CODE_FIPS COUNTY_CODE_FIPS  FIPS
## 1                4               13  4013
## 7               11                1 11001
## 16              13               89 13089
## 18              13              135 13135
## 19              17               31 17031
## 24              23                5 23005
## 26              24                5 24005
## 28              24               27 24027
## 29              24               31 24031
## 30              24               33 24033
## 31              24              510 24510
## 33              25               17 25017
## 35              25               25 25025
## 36              25               27 25027
## 43              27               53 27053
## 49              34               13 34013
## 50              34               17 34017
## 51              34               21 34021
## 52              34               23 34023
## 54              36                1 36001
## 55              36                5 36005
## 57              36               47 36047
## 60              36               61 36061
## 62              36               81 36081
## 63              36               85 36085
## 71              39               35 39035
## 72              39               49 39049
## 79              42               45 42045
## 80              42              101 42101
## 83              48               15 48015
## 86              48              113 48113
## 87              48              157 48157
## 89              48              225 48225
## 90              48              439 48439
## 92              51               59 51059
## 94              51              153 51153
## 96              53               33 53033
```

```r
# Select rows respecting several conditions
data_malaria[data_malaria$STATE == "NEW YORK" & data_malaria$MAL_FREQ_2017 > 10,  ]
```

```
##          COUNTRY    STATE   COUNTY MAL_FREQ_2017 STATE_CODE_FIPS
## 54 UNITED STATES NEW YORK   ALBANY            11              36
## 55 UNITED STATES NEW YORK    BRONX            91              36
## 57 UNITED STATES NEW YORK    KINGS            50              36
## 60 UNITED STATES NEW YORK NEW YORK            42              36
```

```
## 62 UNITED STATES NEW YORK   QUEENS            33             36
## 63 UNITED STATES NEW YORK RICHMOND            11             36
##    COUNTY_CODE_FIPS  FIPS
## 54                1 36001
## 55                5 36005
## 57               47 36047
## 60               61 36061
## 62               81 36081
## 63               85 36085
```

3.1 Activity

- What does the `data_malaria[data_malaria$MAL_FREQ_2017 > 10,  ]` command do?
- Decompose each element of this command:
  - What does `data_malaria$MAL_FREQ_2017` do?
  - What does `data_malaria$MAL_FREQ_2017 < 10` do?
  - What does `data_malaria[...,  ]` do?
- Subset the data set for the data from Colorado only.
- What does the `data_malaria[data_malaria$STATE == "NEW YORK" | data_malaria$STATE == "CALIFORNIA",  ]` command do?

4 Creating New Variable

New variables can be added to existing data frames with `$variable` or `df[ , "variable"]` (the same way existing variables are called). This is also a way to re-code variables (without erasing the original variable).

```
# Create a new variable
data_malaria$YEAR = 2017

# Attribute values based on conditions
data_malaria[data_malaria$MAL_FREQ_2017 <= 10, "MAL_FREQ_QUAL" ] = "LOW"
data_malaria[data_malaria$MAL_FREQ_2017 > 10, "MAL_FREQ_QUAL" ] = "HIGH"
```

4.1 Activity

- Re-code malaria frequency as low (10 or less), high (between 10 and 80), very high (80 or more).

5 Creating Data Frames from Scratch

vectors can be created `c()`, and combined with `cbind()`, `rbind()`, or `data.frame()`.

```
# Counties
COUNTY = data_malaria[data_malaria$STATE == "NEW YORK", "COUNTY"]
length(COUNTY)
```

```
## [1] 12
```

```r
# Create a new variable
DATA_2018 = c(10, 12, 52, 36, 85, 85, 23, 12, 41, 120, 10, 22)
# Or randomly select numbers
DATA_2019 = sample(length(COUNTY))

# Put the data in a data frame
data_malaria_fictive = data.frame(COUNTY = COUNTY,
                                  DATA_2018 = DATA_2018,
                                  DATA_2019 = DATA_2019)
```

5.1 Activity

- Add DATA_2020 to data_malaria_fictive.

6 Pivot

You can pivot tables for the wide to the long format and vice versa.

6.1 Activity

- Pivot data_malaria_fictive from wide to long. You can use AI to help you.

7 Solutions to the activities

```r
# Exploring Data ####

# Calculate the mean and standard deviation of malaria cases for each state
aggregate(MAL_FREQ_2017 ~ STATE, data_malaria, mean)
```

```
##                   STATE MAL_FREQ_2017
## 1               ARIZONA     20.000000
## 2              COLORADO      7.000000
## 3           CONNECTICUT      7.333333
## 4  DISTRICT OF COLUMBIA     19.000000
## 5               FLORIDA      6.833333
## 6               GEORGIA     11.800000
## 7              ILLINOIS     29.000000
## 8               INDIANA      9.000000
## 9                  IOWA      7.000000
## 10               KANSAS      5.500000
## 11                MAINE     11.000000
## 12             MARYLAND     33.857143
## 13        MASSACHUSETTS     16.000000
## 14             MICHIGAN      6.800000
## 15            MINNESOTA     14.666667
## 16             MISSOURI      5.000000
## 17        NEW HAMPSHIRE      6.000000
```

```
## 18           NEW JERSEY     14.714286
## 19             NEW YORK     23.333333
## 20       NORTH CAROLINA      8.250000
## 21         NORTH DAKOTA      6.000000
## 22                 OHIO     11.000000
## 23               OREGON      7.000000
## 24         PENNSYLVANIA     13.600000
## 25         RHODE ISLAND     10.000000
## 26         SOUTH DAKOTA      6.000000
## 27                TEXAS     15.375000
## 28             VIRGINIA     12.200000
## 29           WASHINGTON     13.500000
```

```r
aggregate(MAL_FREQ_2017 ~ STATE, data_malaria, sd)
```

```
##                   STATE MAL_FREQ_2017
## 1               ARIZONA            NA
## 2              COLORADO     1.4142136
## 3           CONNECTICUT     2.5166115
## 4  DISTRICT OF COLUMBIA            NA
## 5               FLORIDA     1.6020820
## 6               GEORGIA     7.5630682
## 7              ILLINOIS            NA
## 8               INDIANA            NA
## 9                  IOWA            NA
## 10               KANSAS     0.7071068
## 11                MAINE            NA
## 12             MARYLAND    36.7850901
## 13        MASSACHUSETTS    12.1655251
## 14             MICHIGAN     1.7888544
## 15            MINNESOTA    15.8850034
## 16             MISSOURI            NA
## 17        NEW HAMPSHIRE            NA
## 18           NEW JERSEY    10.5785047
## 19             NEW YORK    26.3760681
## 20       NORTH CAROLINA     1.7078251
## 21         NORTH DAKOTA            NA
## 22                 OHIO     5.9441848
## 23               OREGON            NA
## 24         PENNSYLVANIA    11.0589330
## 25         RHODE ISLAND            NA
## 26         SOUTH DAKOTA            NA
## 27                TEXAS     9.5907321
## 28             VIRGINIA     7.3620649
## 29           WASHINGTON    10.6066017
```

```r
# Create a new data frame containing the information of the number of cases by state
data_malaria_state = aggregate(MAL_FREQ_2017 ~ STATE, data_malaria, sum)

# Subsetting Data ####

# What does `data_malaria$MAL_FREQ_2017 > 10` do?
data_malaria$MAL_FREQ_2017 > 10
```

```
##  [1]  TRUE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE
## [13] FALSE FALSE FALSE  TRUE FALSE  TRUE  TRUE FALSE FALSE FALSE FALSE  TRUE
## [25] FALSE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE FALSE  TRUE  TRUE
## [37] FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE
## [49]  TRUE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE FALSE  TRUE FALSE FALSE  TRUE
## [61] FALSE  TRUE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE  TRUE
## [73] FALSE FALSE FALSE FALSE FALSE FALSE  TRUE  TRUE FALSE FALSE  TRUE FALSE
## [85] FALSE  TRUE  TRUE FALSE  TRUE  TRUE FALSE  TRUE FALSE  TRUE FALSE  TRUE
## [97] FALSE
```

```r
# Test for each county is the number of malaria case is above 10

# What does `data_malaria[...,  ]` do?
data_malaria[data_malaria$MAL_FREQ_2017 > 10, ]
```

```
##          COUNTRY                STATE               COUNTY MAL_FREQ_2017
## 1  UNITED STATES              ARIZONA             MARICOPA            20
## 7  UNITED STATES DISTRICT OF COLUMBIA DISTRICT OF COLUMBIA            19
## 16 UNITED STATES              GEORGIA               DEKALB            20
## 18 UNITED STATES              GEORGIA             GWINNETT            20
## 19 UNITED STATES             ILLINOIS                 COOK            29
## 24 UNITED STATES                MAINE           CUMBERLAND            11
## 26 UNITED STATES             MARYLAND            BALTIMORE            23
## 28 UNITED STATES             MARYLAND               HOWARD            11
## 29 UNITED STATES             MARYLAND           MONTGOMERY            70
## 30 UNITED STATES             MARYLAND        PRINCE GEORGE           101
## 31 UNITED STATES             MARYLAND       BALTIMORE CITY            16
## 33 UNITED STATES        MASSACHUSETTS            MIDDLESEX            34
## 35 UNITED STATES        MASSACHUSETTS              SUFFOLK            15
## 36 UNITED STATES        MASSACHUSETTS            WORCESTER            21
## 43 UNITED STATES            MINNESOTA             HENNEPIN            33
## 49 UNITED STATES           NEW JERSEY                ESSEX            37
## 50 UNITED STATES           NEW JERSEY               HUDSON            12
## 51 UNITED STATES           NEW JERSEY               MERCER            11
## 52 UNITED STATES           NEW JERSEY            MIDDLESEX            19
## 54 UNITED STATES             NEW YORK               ALBANY            11
## 55 UNITED STATES             NEW YORK                BRONX            91
## 57 UNITED STATES             NEW YORK                KINGS            50
## 60 UNITED STATES             NEW YORK             NEW YORK            42
## 62 UNITED STATES             NEW YORK               QUEENS            33
## 63 UNITED STATES             NEW YORK             RICHMOND            11
## 71 UNITED STATES                 OHIO             CUYAHOGA            12
## 72 UNITED STATES                 OHIO             FRANKLIN            19
## 79 UNITED STATES         PENNSYLVANIA             DELAWARE            20
## 80 UNITED STATES         PENNSYLVANIA         PHILADELPHIA            30
## 83 UNITED STATES                TEXAS               AUSTIN            12
## 86 UNITED STATES                TEXAS               DALLAS            19
## 87 UNITED STATES                TEXAS            FORT BEND            17
## 89 UNITED STATES                TEXAS              HOUSTON            35
## 90 UNITED STATES                TEXAS              TARRANT            19
## 92 UNITED STATES             VIRGINIA              FAIRFAX            24
## 94 UNITED STATES             VIRGINIA       PRINCE WILLIAM            14
## 96 UNITED STATES           WASHINGTON                 KING            21
##    STATE_CODE_FIPS COUNTY_CODE_FIPS  FIPS YEAR MAL_FREQ_QUAL
```

```
## 1                4               13  4013 2017          HIGH
## 7               11                1 11001 2017          HIGH
## 16              13               89 13089 2017          HIGH
## 18              13              135 13135 2017          HIGH
## 19              17               31 17031 2017          HIGH
## 24              23                5 23005 2017          HIGH
## 26              24                5 24005 2017          HIGH
## 28              24               27 24027 2017          HIGH
## 29              24               31 24031 2017          HIGH
## 30              24               33 24033 2017          HIGH
## 31              24              510 24510 2017          HIGH
## 33              25               17 25017 2017          HIGH
## 35              25               25 25025 2017          HIGH
## 36              25               27 25027 2017          HIGH
## 43              27               53 27053 2017          HIGH
## 49              34               13 34013 2017          HIGH
## 50              34               17 34017 2017          HIGH
## 51              34               21 34021 2017          HIGH
## 52              34               23 34023 2017          HIGH
## 54              36                1 36001 2017          HIGH
## 55              36                5 36005 2017          HIGH
## 57              36               47 36047 2017          HIGH
## 60              36               61 36061 2017          HIGH
## 62              36               81 36081 2017          HIGH
## 63              36               85 36085 2017          HIGH
## 71              39               35 39035 2017          HIGH
## 72              39               49 39049 2017          HIGH
## 79              42               45 42045 2017          HIGH
## 80              42              101 42101 2017          HIGH
## 83              48               15 48015 2017          HIGH
## 86              48              113 48113 2017          HIGH
## 87              48              157 48157 2017          HIGH
## 89              48              225 48225 2017          HIGH
## 90              48              439 48439 2017          HIGH
## 92              51               59 51059 2017          HIGH
## 94              51              153 51153 2017          HIGH
## 96              53               33 53033 2017          HIGH
```

```r
# Call the rows that verify the condition # cases > 10, and all the columns

# Subset the data set for the data from Colorado only.
data_malaria[data_malaria$STATE == "COLORADO", ]
```

```
##         COUNTRY    STATE   COUNTY MAL_FREQ_2017 STATE_CODE_FIPS
## 2 UNITED STATES COLORADO ARAPAHOE             8               8
## 3 UNITED STATES COLORADO   DENVER             6               8
##   COUNTY_CODE_FIPS FIPS YEAR MAL_FREQ_QUAL
## 2                5 8005 2017           LOW
## 3               31 8031 2017           LOW
```

```r
# Creating New Variables ####

# Re-code malaria frequency as low (10 or less), high (between 10 and 80), very high (80 or more).
```

```
data_malaria[data_malaria$MAL_FREQ_2017 <= 10, "MAL_FREQ_QUAL" ] = "LOW"
data_malaria[data_malaria$MAL_FREQ_2017 > 10 &
               data_malaria$MAL_FREQ_2017 < 80,
             "MAL_FREQ_QUAL" ] = "HIGH"
data_malaria[data_malaria$MAL_FREQ_2017 >= 80, "MAL_FREQ_QUAL" ] = "VERY HIGH"

# Creating Data Frames from Scratch ####

# Add DATA_2020 to data_malaria_fictive.
data_malaria_fictive$DATA_2020 = sample(nrow(data_malaria_fictive))

# Pivot ####

# Pivot data_malaria_fictive from wide to long. You can use AI to help you.
# Indicate to AI:
# - The language (R)
# - The task (pivot from wide to long)
# - The names of your variables (from colnames(data_malaria_fictive))
# - Adjust the code (e.g., rename object if you need/want)
# - Always check!

colnames(data_malaria_fictive)
```

```
## [1] "COUNTY"    "DATA_2018" "DATA_2019" "DATA_2020"
```

```
# Prompt example:
# In R, I have a data frame with these variables: "COUNTY"    "DATA_2018" "DATA_2019" "DATA_2020"
# I want to pivot it from wide to long

library(tidyverse)

# Pivot to long format
data_malaria_fictive_long <- data_malaria_fictive %>%
  pivot_longer(cols = starts_with("DATA_"),
               names_to = "year",
               values_to = "cases")

# Clean up the year column (remove "DATA_" prefix)
data_malaria_fictive_long <- data_malaria_fictive_long %>%
  mutate(year = str_remove(year, "DATA_"))

print(data_malaria_fictive_long)
```

```
## # A tibble: 36 x 3
##    COUNTY year  cases
##    <chr>  <chr> <dbl>
##  1 ALBANY 2018     10
##  2 ALBANY 2019     10
##  3 ALBANY 2020      1
##  4 BRONX  2018     12
##  5 BRONX  2019      6
##  6 BRONX  2020      6
##  7 ERIE   2018     52
```

```
## 8 ERIE   2019      8
## 9 ERIE   2020      7
## 10 KINGS 2018     36
## # i 26 more rows
```

The End.